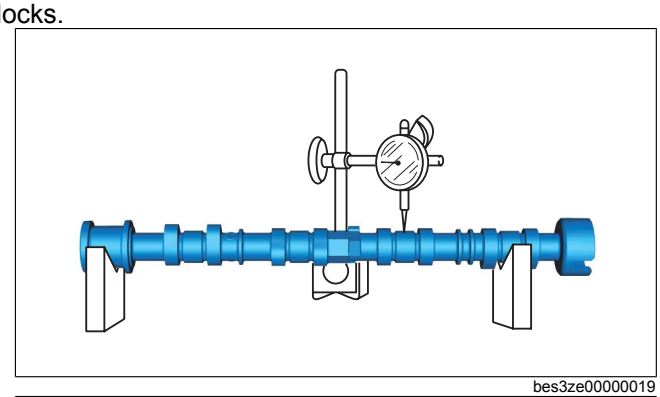
CAMSHAFT INSPECTION

id011000501500

1. Set the No.1 and No.5 journals of the camshaft on V-blocks.
2. Measure the camshaft runout using the dial gauge.
   - If it exceeds the maximum specification, replace the camshaft.

   Maximum camshaft runout
   0.030 mm {0.0012 in}



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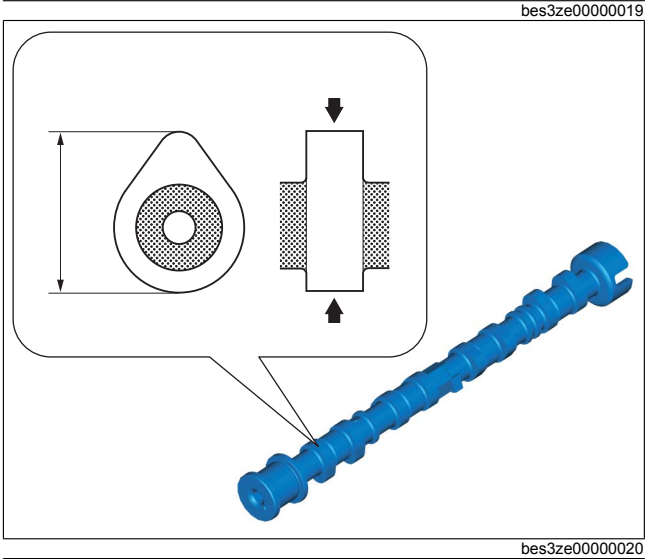
3. Measure the cam height using the micrometer.
   - If it is less than the minimum specification, replace the camshaft.

   Standard cam height [intake camshaft]
   34.12 mm {1.343 in}

   Standard cam height [exhaust camshaft]
   34.48 mm {1.357 in}

   Minimum cam height [intake camshaft]
   34.05 mm {1.341 in}

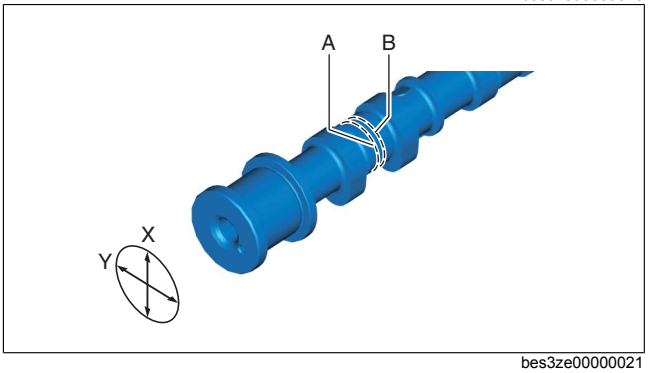
   Minimum cam height [exhaust camshaft]
   34.41 mm {1.355 in}



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4. Measure the journal diameter using the micrometer. Measurement positions total four and are in the X and Y directions, at two points (A and B) as shown in the figure.
   - If it is less than the minimum specification, replace the camshaft.

   Note
   - The journal diameter of only the exhaust camshaft No.1 journal differs.



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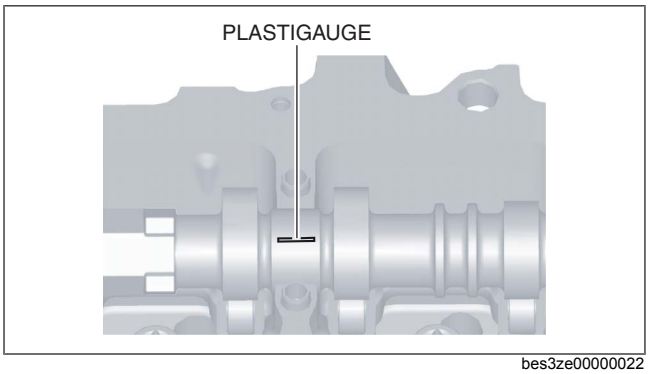
   Standard camshaft journal diameter [intake camshaft]
   No.1 journal: 33.950—33.970 mm {1.3367—1.3374 in}
   Except No.1 journal: 25.950—25.975 mm {1.0217—1.0226 in}

   Standard camshaft journal diameter [exhaust camshaft]
   No.1 journal: 33.950—33.975 mm {1.3367—1.3375 in}
   Except No.1 journal: 25.950—25.975 mm {1.0217—1.0226 in}

   Minimum camshaft journal diameter
   No.1 journal: 33.920 mm {1.3354 in}
   Except No.1 journal: 25.920 mm {1.0205 in}

5. Measure the camshaft journal oil clearance using the following procedure:
   (1) Clean the camshaft journal and the journal receptacle part.
   (2) Put the camshaft on the cylinder head with the rocker arm detached.

(3) Cut the plastigauge to the same length as the journal width and position it parallel to the camshaft.
(4) Install the camshaft caps. (See CYLINDER HEAD ASSEMBLY (II).)
(5) Remove the camshaft caps. (See CYLINDER HEAD DISASSEMBLY (I).)
(6) Measure the camshaft journal oil clearance.
    • If it exceeds the maximum specification, replace the cylinder head.



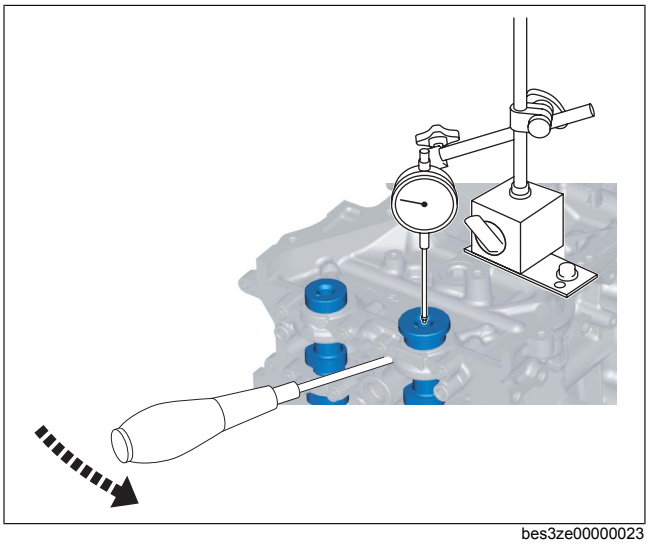
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Standard camshaft journal oil clearance [intake camshaft]
No.1 journal: 0.030—0.075 mm {0.0012—0.0029 in}
Except No.1 journal: 0.025—0.071 mm {0.0010—0.0027 in}

Standard camshaft journal oil clearance [exhaust camshaft]
No.1 journal: 0.025—0.075 mm {0.0010—0.0029 in}
Except No.1 journal: 0.025—0.071 mm {0.0010—0.0027 in}

Maximum camshaft journal oil clearance
No.1 journal: 0.105 mm {0.00413 in}
Except No.1 journal: 0.101 mm {0.00398 in}

6. Measure the camshaft end play using a dial gauge.
    • If it exceeds the maximum specification, replace the cylinder head or camshaft.


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Standard camshaft end play
0.07—0.22 mm {0.003—0.008 in}

Maximum camshaft end play
0.98 mm {0.039 in}